

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The meeting focused on three primary topics: privacy pools, understanding as key, and biometrics for key management. The discussion explored the technical and regulatory implications of these technologies in blockchain applications. Key points included the need to incentivize institutions to accept zero-knowledge proofs, the use of proverb compressions for secure communication, and the challenges and solutions related to using biometric data for wallet generation and recovery. The session also highlighted the importance of aligning with TradFi practices, ensuring user choice, and addressing regulatory requirements.

Key Discussion Points

1. Privacy Pools

- **Incentivizing Institutions:** The discussion centered on how to incentivize institutions to accept zero-knowledge proofs (ZKPs) for compliance purposes. Participants emphasized the importance of providing clear guidelines and incentives to facilitate adoption.
- **RFP from Ethereum:** Feedback from the Ethereum RFP was shared, highlighting the need for further clarification of institutional acceptance criteria. The group agreed to explore potential TradFi customs and practices that can be integrated into the project.

2. Understanding as Key

- **Proverb Compressions:** Keegan presented research on using proverb compressions as a key ceremony for secure communication through encrypted channels on-chain. Participants discussed the potential of sharing proverbs selectively to enhance security.
- **Selective Disclosure:** The mechanism of selective disclosure was explained, where one party can choose to reveal information to the transparent chain without exposing sensitive details.

3. Biometrics for Key Management

- **Backup Requirements:** The necessity and challenges of having a backup for biometric data were discussed, especially in scenarios where biometric sources are compromised. Participants agreed that while biometrics offer convenience, they require robust backup solutions.

- **Entropy and Noise:** Detailed insights were provided on measuring entropy and handling noise in biometric data. Different modalities like face, finger, and iris were compared, with a focus on the number of errors and noise levels.
- **Perception and Regulation:** The session addressed the pushback from the crypto community regarding privacy concerns and regulatory requirements for making biometric use optional. Participants emphasized the importance of user choice and compliance.

Action Items and Next Steps

1. Privacy Pools

- Conduct further research on incentivizing institutions to accept zero-knowledge proofs.
- Engage with Ethereum RFP stakeholders to clarify institutional acceptance criteria.
- Explore potential TradFi customs and practices that can be integrated into the project.

2. Understanding as Key

- Develop a detailed protocol for using proverb compressions in key ceremonies.
- Test the selective disclosure mechanism on various blockchain platforms.
- Document best practices and use cases for secure communication.

3. Biometrics for Key Management

- Investigate advanced biometric techniques to improve reliability and security.
- Collaborate with biometric technology providers like Hitachi and NEC to explore integration possibilities.
- Engage with regulatory bodies to ensure compliance with optional biometric usage requirements.

Detailed Session Summary

Privacy Pools - Incentivizing Institutions: The session began with a discussion on how institutions can be incentivized to accept zero-knowledge proofs (ZKPs) for compliance purposes. Participants emphasized the importance of aligning with TradFi customs and practices to gain institutional trust. - **RFP from Ethereum:** Feedback from the Ethereum RFP was shared, highlighting the need for clear compliance criteria and institutional acceptance. The group agreed to further explore these areas to ensure project success.

Understanding as Key - Proverb Compressions: Keegan presented his research on using proverb compressions as a key ceremony for secure communication.

Participants discussed the potential of sharing proverbs through encrypted channels on-chain and selectively disclosing them to the transparent chain. - **Selective**

Disclosure: The mechanism of selective disclosure was explained, where one party can choose to reveal information to the transparent chain without exposing sensitive details.

Biometrics for Key Management - Backup Requirements: The discussion covered the necessity of having a backup for biometric data, especially in scenarios where biometric sources are compromised. Participants agreed that while biometrics offer convenience, they require robust backup solutions. - **Entropy and Noise:** Detailed

insights were provided on measuring entropy and handling noise in biometric data. Different modalities like face, finger, and iris were compared, with a focus on the

number of errors and noise levels. - **Perception and Regulation:** The session addressed the pushback from the crypto community regarding privacy concerns and regulatory requirements for making biometric use optional. Participants emphasized the importance of user choice and compliance.

Closing Remarks - Keegan summarized the key insights and feedback received during the session, expressing gratitude for the valuable discussions. - The group agreed to continue exploring these topics in future sessions and to document best practices and use cases. - The next topic for Block 14 will be forensic analysis towards a common lexicon of harmful on-chain activities.

Appendix: Full Meeting Transcript (Anonymized)

28.6 --> 30.0: This is a meeting transcript.

58.6 --> 60.0: This is a meeting transcript.

69.6 --> 77.3: This is a meeting transcript.

90.0 --> 118.8: I think what we should do is just like get started with your presentation, and I've been thinking a lot about the privacy pools improvement paper that we've been presenting and working on in a different context that I'd like to bring to the conversation, but there is also, not here at the moment, but there may have been another person who was going to present on wallet stuff.

119.4 --> 121.5: But yeah, Keegan, you should just get started.

122.5 --> 128.8: All right. So, I don't see anyone on the roundtable, so can I assume everyone see my screen anyway?

130.1 --> 132.0: You're not sharing yet, but...

132.0 --> 134.9: I'm going to share it now, right now, but...

134.9 --> 135.7: Yeah, we can see the screen.

136.3 --> 138.8: Yeah, I don't know how the room is set up, so...

139.8 --> 140.2: Okay.
141.6 --> 143.2: Let me turn on the screen sharing.
147.9 --> 151.9: And hopefully... Oh, no, I think you guys see the present review.
153.1 --> 153.7: Yeah.
155.7 --> 156.8: Let me try again.
160.5 --> 163.5: What about this? All right.
165.8 --> 166.6: Is it working?
168.2 --> 169.0: Seems working.
169.3 --> 170.4: Yeah, we got you.
171.1 --> 172.2: Okay, so can I start now?
173.2 --> 173.9: Go for it.
174.2 --> 174.8: All right.
176.2 --> 188.7: So, thank you for coming to this session, and let me kick off this session by giving the presentation about the project titled Accountable Wallet for Privacy Pools Perfection.
189.2 --> 194.7: I'm Keegan Fukuda from Virgin Attack, and this is going to be maybe a 20-minute-ish presentation from me.
197.0 --> 204.6: So, let me first explain what the Privacy Pools is, in case any folks in the room are not familiar with this protocol.
205.1 --> 209.9: So, the Privacy Pool is a protocol to balance the compliance and the on-chain privacy.
210.5 --> 219.8: It's proposed in 2023 by Bitlyk, and implemented in 2025 by Oxbow in July.
219.8 --> 232.6: And as it states, I mean, instead of merely Zillow is proving the link between withdrawals and deposit, the user also proved membership in an associational set, like set of compliant deposit.
233.4 --> 238.8: And I don't know if this sentence makes any sense, so I got this on visualization.
238.8 --> 250.5: So, basically, the Privacy Pool stemmed from the other protocol called TornadoCash, which is the privacy-enhancing solution.
251.0 --> 256.3: And it got sanctioned by the US Department of the Treasury in 2022, which is lifted in 2025.
256.3 --> 268.3: But what the TornadoCash will do is, as you guys can see on this figure, wallets are making deposits first and make withdrawals after that, later on.
268.9 --> 277.1: And what TornadoCash will do is cutting the link between this deposit and the withdrawals, using the terminology proof technologies.
277.8 --> 283.0: So, this deposit and the withdrawals got unlinkable from anyone else.
284.2 --> 293.4: But this got used by some malicious actors like terrorists for money laundering purposes.
294.1 --> 296.1: And that's why it got sanctioned once.
297.0 --> 307.3: And as a response against that momentum, the Privacy Pool is proposed as an equilibrium between the privacy and compliance.
307.3 --> 322.4: And built on top of the TornadoCash, what it will do is,

there is a set of deposits called association set, which are believed to be compliant.

322.4 --> 338.2: And any withdrawal coming from the deposit included in the association set can also prove that the withdrawals came from any deposit in the association set.

338.6 --> 350.2: While prohibiting any withdrawals coming from non-association set deposit, forging proof of the withdrawal from the association set.

350.2 --> 359.1: By doing so, it can separate the compliant deposit and uncompliant deposit by defining the association set.

359.9 --> 375.5: And in practice, the entity called association set provider is assumed, which generates the association set that has certain properties like meeting the compliant requirements.

377.7 --> 380.6: And so what's missing in Privacy Pools?

383.7 --> 387.2: In this session, I think we are covering two missing parts.

387.3 --> 392.6: The first one is the centralized ASP, which is what this project is actually trying to solve.

393.1 --> 395.5: And also the second one is the institutional integration.

396.4 --> 401.2: This is mentioned in the RFP recently published by Ethereum Foundation.

401.2 --> 409.9: And I think this is going to be covered later in the session, hopefully.

411.3 --> 413.9: So let me focus on the centralized ASP problem.

414.5 --> 416.8: So what is bad with the centralized ASP?

418.9 --> 431.0: In this setting, the malicious ASP can restrict users from depositing coin into its association set, giving some special privilege to specific entities.

431.4 --> 445.9: Also, malicious ASP can compile the association set in a way that allows them to maximize the extractable information or inflate the perceived anonymity by adding some deposit for corresponding withdrawals or none or something like that.

445.9 --> 455.4: Anyway, in short, this centralized ASP problem is making the critical single point of failure.

457.5 --> 461.1: Let me cover the original Accountable Wallet idea here.

461.6 --> 468.6: It is proposed in the BEGIN and FC workshop two or three years ago.

469.8 --> 482.4: In this original idea, the wallets issue proof of the legitimacy of the wallet owners and also the addresses and also the coin asset, I mean, the coin provenance of the wallet.

483.2 --> 487.4: And those proofs will be propagated along with the transaction.

487.4 --> 507.6: And so that it's like forming a chain of proof and also by aggregating them, we can calculate the trust score so that it's aiming to ensure the accountability in the peer-to-peer transaction space and also incentivize the good conduct by clearly assigning the score of each

wallet.

508.6 --> 522.4: And I'm not going to cover details of it, but five or six proofs are intricately making that chain.

522.4 --> 529.0: And this kind of remained like concept level.

529.8 --> 548.4: And my work was actually designing the protocol to achieve this goal and also simplify it to be simple enough to make some implementation and also add some security assessment.

548.4 --> 566.9: And, okay, so it got kind of long, but anyway, so the Accountable Wallet in short is like user record, prove and verify their own chain compliance and eventually form an association set, which is like becoming like a community curation of the association set.

566.9 --> 580.7: And to simplify it, I break it down into three components. At the most fundamental level, I designed a protocol called Verifiable Compliance Affidavit.

581.6 --> 592.6: And this portion, this component, I mean, this protocol is submitted to the IEEE ICPC, which is the top conference, one of the top conferences in the blockchain field.

592.6 --> 604.1: It's under review, hopefully get through. And on top of this Affidavit protocol, we can build a compliance scoring mechanism.

605.0 --> 614.5: And eventually built on these two fundamental components, we can think of decentralized association set construction.

614.5 --> 633.7: And this is the one that solves the centralized ASP. And in this presentation, given the time limitations, I'm going to focus on the first focus on the Verifiable Compliance Affidavit, which is a protocol submitted to the conference.

633.7 --> 643.6: And also later in the presentation, I'm going to cover the compliance scoring mechanism, which is like remain a future work in the paper I submitted.

644.3 --> 654.9: And hopefully, if we can spend some time in the discussion for a concrete design of this scoring mechanism, that would be awesome.

654.9 --> 669.5: And so let me go to the Verifiable Compliance Affidavit. So this is like the simplest version of the Accountable Wallet.

669.5 --> 676.2: So wallets issued proof of due diligence, and it's going to be recorded along with the transaction.

677.3 --> 689.7: And this protocol itself is aiming to mitigate the asymmetry between the institutional risk assessment capabilities and the individual users' capacity for the self-defense.

689.7 --> 698.7: So what it means is like the cryptographic proof can serve as the admissible evidence of a good faith of the user.

699.6 --> 705.1: And that can be used in a rebuttal against the centralized compliance determination.

705.1 --> 719.2: So what it means is like, so the risk assessment at the FASP, for example, and I mean, they're following various regulations, such as like FATF recommendation, US Bank Secrecy Act and Clarity Act.

719.7 --> 732.3: And those regulations are making almost mandatory to use the blockchain analytics tool, which is a black box for end users.

732.3 --> 748.7: And so by recording the proof of the best effort due diligence against the counterparty, at least the user can, at least this protocol will enhance the user's self-defense capacity.

749.4 --> 752.6: So that's what this protocol is aiming, basically.

752.6 --> 762.1: And we call this proof, cryptographic proof of the good faith as a Verifiable Compliance Record, VCR in short.

763.2 --> 769.5: And this is the overview, the snapshot of how the protocol works.

769.5 --> 773.5: And there are a lot of things to cover here, actually.

773.9 --> 784.2: But what is important is like there is sender and receiver and also a crypto watch list, which is the list of the sanctioned addresses.

784.8 --> 786.6: And there could be multiple watch lists.

787.6 --> 800.4: So what's happened is, before sender incurring the transaction, the receiver will conduct the off-chain due diligence following their own policy.

800.8 --> 806.2: This policy means like which crypto watch list this receiver will refer in the due diligence.

806.2 --> 819.8: Like, for example, if this receiver lives in Japan, then he will likely refer the crypto watch list issued by the Japanese government agencies like financial service agencies.

819.8 --> 837.2: And within the off-chain due diligence, the receiver will check if the sender's address is absent from the crypto watch list or not.

837.2 --> 842.8: And if it's absent, then the receiver will generate the non-membership proof.

843.8 --> 850.6: And with some metadata, this data package will be signed by the receiver and sent to the sender.

851.1 --> 856.9: And the sender will record this data pack with the digital signature to the blockchain.

857.3 --> 860.5: And then this data pack is called VCR on-chain.

863.9 --> 867.9: And this VCR will be recorded along with the transaction.

868.3 --> 873.8: And any third party verifier can verify this VCR at any time they want.

874.5 --> 885.3: So in summary, the receiver checks the sender's absence from the chosen crypto watch list and records the non-membership proof on-chain storage with the signature.

886.3 --> 891.2: This is the simplified version of the compliance protocol.

891.9 --> 897.8: And in the paper, I defined some security requirements and wrote some proof.

898.0 --> 914.3: And also I did some implementation for verifying the performance of this protocol for generating, also verifying the VCR, how the VCR size will be, and also what's the estimated gas cost for recording it.

914.3 --> 926.8: And built on top of this verifiable compliance affidavit

t, now we can move on to the scoring mechanism.

927.5 --> 931.0: I think it's kind of running out of time, so let me go kind of quickly.

934.5 --> 950.2: Anyway, what this protocol verifiable compliance affidavit is doing is the recipient checks the sender's absence from the crypto watch list at the time they transact. That's all.

951.2 --> 956.3: But practically, we want to consider the coin provenance in a due diligence.

957.3 --> 963.9: So let us define the provenance score as the simplest form of the provenance compliance level.

964.6 --> 971.5: Like this sender has the score 0.75 or 0.9, which is supposed to be more than average.

971.9 --> 975.3: That means the provenance seems good, something like that.

975.3 --> 982.7: And also this, I'm not going to cover the detail of the definition of the score.

983.7 --> 988.0: This score calculation should have something to do with this VCR.

988.2 --> 999.2: For example, if the transaction has VCR, then the upstream at the sender's score will be propagated directly to the receiver, something like that.

999.2 --> 1010.4: Anyway, this provenance definition will come from how compliant the wallet owner behaves.

1010.7 --> 1023.4: That means how much, for example, it's going to reflect the portion of the transaction that has the VCR along with them, for example.

1027.8 --> 1040.8: Anyway, from this recipient side, to verify the compliance score of this sender, that's going to incur the exponential verification cost.

1040.8 --> 1046.1: Because this sender's score is calculated based on the upstream sender's score.

1046.9 --> 1055.6: And this upstream sender's score will be also calculated based on this further upstream discontinuous growing,

1056.4 --> 1065.7: until we get to the trusted deposit or any sanctioned deposit or whatever, like some fixed score entity.

1065.7 --> 1071.4: And so we need to mitigate this exponential verification cost.

1072.3 --> 1078.0: And so that's why we are employing some third-party auditing entity called watchers.

1078.2 --> 1084.9: So these watchers are responsible for the real-time verification of the VCR as they are broadcast to the blockchain.

1085.6 --> 1089.0: So therefore, they keep track of the provenance score for each wallet.

1090.0 --> 1096.4: And each watcher operates according to its own verified policy.

1096.7 --> 1103.0: So that means this watcher will refer to this CryptWatch list, but the other watcher will refer to something else.

1103.6 --> 1112.1: So that the score is not universal, but based on what the watcher chooses as a policy, the score will be defined based on that.

1113.6 --> 1118.1: So the recipient doesn't really have to go back to the transaction history,

1118.3 --> 1130.4: but the watcher will ensure the correctness of the provenance score of the sender.

1132.3 --> 1136.1: So I come up with two watcher operational models.

1136.4 --> 1137.8: One is the query-based model.

1137.8 --> 1143.2: So the watcher maintains a database that tracks the compliance score of each wallet addresses

1143.2 --> 1148.7: and the receiver queries the watchers to obtain the sender's current provenance score.

1149.3 --> 1153.9: This is operationally simple, but it's going to introduce the watcher as a single point of failure.

1154.2 --> 1163.4: In case the watcher is out of service, then no user will be able to refer the score at that time.

1163.5 --> 1164.4: So it's going to be problematic.

1164.4 --> 1169.2: And the second model is the attestation-based model.

1170.5 --> 1177.6: In this case, the user will include the locally computed provenance score in VCR

1177.6 --> 1181.1: and the watchers validate the correctness of the score computation.

1181.9 --> 1187.6: And users can still calculate scores as long as the watcher has verified upstream VCRs.

1188.3 --> 1193.1: So this model will mitigate the watcher becoming the single point of failure,

1193.1 --> 1195.6: but this requires some additional mechanism,

1195.9 --> 1202.8: for example, issuing the corrective attestation to offset any fraudulent VCR

1202.8 --> 1207.6: that includes some fake scores or something like that.

1215.1 --> 1219.9: But what's important is this watcher itself,

1219.9 --> 1221.4: with the watcher's operation,

1222.5 --> 1227.5: basically, I mean, they're responsible for a real-time verification of the VCR

1227.5 --> 1233.9: and also keep tracking the record of the compliance score.

1234.4 --> 1241.5: But this compliance score calculation will be only based on the public information.

1242.1 --> 1245.8: So whether this transaction has valid VCR, that's also public.

1245.8 --> 1249.5: And whether this deposit is trustworthy,

1249.7 --> 1254.1: for example, the deposit from the Coinbase, which is supposed to be trustworthy,

1254.4 --> 1255.9: then that's also public.

1256.5 --> 1259.8: And also, obviously, the CryptoWatchList is also public.

1260.2 --> 1262.8: So the score calculation itself is public.

1263.2 --> 1266.5: I mean, it's verifiable by anyone.
1266.8 --> 1268.9: So any malicious operation by a watcher,
1268.9 --> 1275.6: like forging a higher score or damaging someone's score is detectable.
1277.1 --> 1280.8: And so the open question for this floor will be,
1281.1 --> 1283.5: who will be the watcher practically?
1283.8 --> 1285.9: And how can we establish the watcher's trust?
1286.5 --> 1291.4: So even I said, I mean, the watcher fraudulent operation will be detectable.
1292.0 --> 1296.5: That doesn't mean anyone can be the watcher.
1296.5 --> 1301.6: Maybe that could be the model, but still we need to discuss who will be the watcher anyway.
1302.7 --> 1309.4: And also the integration with the privacy pool as the way to construct the association set
1309.4 --> 1313.5: will be also worth to discuss.
1314.1 --> 1319.1: And also, as the original AccountableWallet suggests,
1319.9 --> 1323.0: the off-chain identity integration is also important practically.
1324.2 --> 1328.6: And also its potential privacy impacts, that is also important.
1329.3 --> 1338.3: And also the risk of metadata leakage through recording the VCR on-chain is also worth to consider.
1338.5 --> 1349.8: Because if one user is continuously using the specific CryptoWatchList,
1350.1 --> 1352.1: which is issued by Japanese government,
1352.1 --> 1356.4: then that's going to almost revealing that the wallet owner lives in Japan.
1357.3 --> 1360.9: And that will be the potential privacy impact.
1361.8 --> 1366.5: And this dusting attack resistance, that's also important.
1366.9 --> 1378.7: Like some malicious user send dirty coin to some honest user to damage his provenance score.
1378.7 --> 1385.4: That could be one of the major attacks that we can think of.
1385.6 --> 1388.2: And the redemption mechanism should be designed.
1389.1 --> 1396.0: And lastly, how does our proposal resonate with the problem defined in the RFP,
1396.3 --> 1398.7: Privacy Pools Institutional Integration?
1399.5 --> 1401.0: And what's the meaning of institutional?
1401.0 --> 1412.2: Basically, what's the definition of the second missing part in privacy pool, like I covered here?
1413.2 --> 1415.2: That's actually my personal interest.
1415.7 --> 1420.0: And if I can find any answer or clue in this session, that'll be appreciated.
1420.8 --> 1422.2: And that's all from me. Thank you so much.
1424.0 --> 1424.8: Thank you.
1425.9 --> 1430.3: It's really great to see the progression of this research.

arch.

1431.3 --> 1440.6: The things that I've got written down at the moment, Keegan, is one quick just semantic check.

1441.0 --> 1450.1: VCR is Verifiable Credential Registry in DID and VC 1 and.

1450.8 --> 1451.3: So there may be...

1453.1 --> 1455.9: Do you mean VDR? I forgot.

1455.9 --> 1460.0: No, I'm talking about like a separate ecosystem.

1460.8 --> 1466.7: Verifiable Compliance Record uses the same acronym as Verifiable Credential Registry.

1467.5 --> 1471.1: And I think usually these acronyms, it's not a problem.

1471.8 --> 1476.6: But I think that a Verifiable Credential Registry is actually an association set.

1477.1 --> 1478.4: So it could be confusing.

1480.2 --> 1484.1: Whether or not you're dealing with the registry or you're dealing with a single record.

1484.8 --> 1493.1: And then my answer to that question is, why not just VCA and be more specific and say that it is the Verifiable Compliance Affidavit?

1494.1 --> 1497.9: Because the word affidavit here, I think, is the most important one.

1500.1 --> 1501.4: Well, yeah, that's true.

1502.0 --> 1509.8: I use record because this proof can be used as the source of the score calculation.

1510.1 --> 1513.1: And that's not affidavit we're supposed to do.

1513.1 --> 1515.9: I mean, I'm not very sure about the meaning.

1516.4 --> 1526.5: I mean, the appropriate way to use the word affidavit, but I get your point.

1527.3 --> 1529.1: I like that it's a legal term.

1529.8 --> 1530.6: Right, it's a legal term.

1531.0 --> 1535.1: And I don't know if it's appropriate to use that for the scoring stuff.

1536.1 --> 1537.6: But I get your point.

1537.7 --> 1538.3: Thank you so much.

1538.3 --> 1560.5: Yeah, and then you had on the questions page, one thing that I think is going to be important here, and this is for the association set provider side of things, is around, there's an organization called Glianet, which is trying to build net fiduciaries.

1561.0 --> 1562.6: So data net fiduciaries.

1562.6 --> 1567.0: And my assumption is that these watchers are going to be nodes.

1568.2 --> 1572.7: And they need to read and write chain.

1573.2 --> 1586.0: So we should potentially, that seems like a natural path to build watcher trust, even though it starts centralized and hopefully moves decentralized over time and we get more information.

1586.0 --> 1602.2: Is that you could require like a Glianet pledge to buy

ild in the web of trust, which gives different obligations or responsibilities to nodes to be good about the data and a whole bunch of security practices.

1603.0 --> 1605.7: So I think in terms of implementation, we should think about that.

1606.8 --> 1616.2: They were my two direct points about like just specific feedback on the work you've done, which is brilliant.

1616.8 --> 1629.1: The question that I've got for the room to like open the conversation around the privacy pool side of things is, well actually, I've got two questions.

1629.8 --> 1642.8: One of the questions is, and we spoke about this last year as well, is do these privacy pools only need to be moving money or can they also be moving information?

1643.3 --> 1644.7: And that's more of like an open question.

1645.4 --> 1648.5: And I've got a question specifically for you, Keegan.

1649.3 --> 1656.0: Could you go to the page that shows Nisato's research where it shows the chain of proof?

1656.9 --> 1659.4: This one. Wait, give me a sec. Here?

1659.9 --> 1660.7: Yeah, that one.

1661.3 --> 1672.5: One thing that I've been working on at the moment is rather than like proof of observation of a particular state, you try and do like proof of trajectory through the state.

1672.9 --> 1683.4: So you find a point at one point in time and then you prove that you were the only person that could possibly have gone on that path in this future point.

1683.4 --> 1691.9: So wallet A to wallet D, like you don't need wallet B and C in the middle because you have that like chain of proof.

1692.1 --> 1702.5: So you would only need to provide like the origination and then D1 in order to get that legitimacy of trajectory rather than state.

1702.5 --> 1723.6: So that's something that I think we could also improve on this research is like looking into is there a way to make this not just a one or two dimensional graph, but like three dimensional so you can build a trajectory through rather than like a proof of provenance system.

1724.6 --> 1726.8: But yeah, I'm going to pause.

1728.3 --> 1744.8: Well, just for clarification, did you say the privacy pool? I mean, I'm sorry, this graph, the chain of proof can be extended to prove the trajectory, not only about the state, but...

1744.8 --> 1751.0: Yeah, and if you prove that your trajectory was a proof of honest trajectory.

1751.8 --> 1759.3: So I guess the word honest here is important that like my intention when I first got this was to do this.

1759.3 --> 1780.5: And then this is what happened. And we can see that on chain and it ended here. And that forms some sort of selective disclosure about this was the trajectory, which means that we don't need to check every wallet jump in between, because that information is already there, if that makes sense.

1781.3 --> 1783.9: I think that makes sense.

1787.4 --> 1797.5: I don't know. I also think it's important to think of the coexisting of this protocol with the privacy pool. I mean, such like privacy enhancing solutions.

1798.8 --> 1808.5: And from my understanding, those protocol is designed to actually hide the trajectory of the specific transaction by cutting the linkability.

1808.5 --> 1822.7: And so, like what I'm kind of imagining is like, maybe this proof can be substituted by some proof issued by the privacy pool, you know, smart contract or like the other

1825.3 --> 1851.6: implementation, not rather by like some user issuing the proof, so that like maybe this wallet B is saying, instead of this wallet B saying I got this credential or proof from wallet A, maybe this wallet B will say like I got this proof that my withdrawal from this pool has as some sort of like meeting this compliance requirement.

1851.6 --> 1863.7: And like, anyway, belong to this association set, so that without revealing, you know, it comes from the wallet A, I can still prove that like, you know, this trustworthy.

1863.7 --> 1875.9: Exactly. You've caught it. So like the fact that you went through the privacy pool and went through this association set is enough of information to prove that the wallet is legitimate without needing to create the chain of proof outside.

1876.6 --> 1876.7: Yeah.

1877.2 --> 1878.2: Okay, I get your point.

1895.1 --> 1903.3: Has anyone got any thoughts or comments on the institutional question that Keegan had on the last slide?

1904.3 --> 1916.8: Ethereum is interested in institutional approaches to privacy pools or, yeah, what does that mean? What is an institutional privacy pool?

1929.9 --> 1940.6: Maybe we can first share the RFP. So that like, yeah, go for it. The whole room. I mean, I didn't prepare for that. But give me some seconds to get the RFP.

1941.0 --> 1941.3: Yeah.

1957.9 --> 1959.3: Okay.

2000.6 --> 2002.7: Yeah, I'm just going to keep talking.

2002.7 --> 2028.7: This proof of honest trajectory is interesting because if you, it's almost like you're getting privacy by going through the association set, but by proving that you went through that process, you're giving like increased trust to your wallet, because you've like selectively decided to put it through something that checks it against different sanctions and lists.

2030.9 --> 2033.8: It could almost be like a pre.

2036.2 --> 2038.3: Anyway, yeah, just thinking about that.

2039.3 --> 2042.1: So, yeah, privacy pools, institutional integration.

2043.4 --> 2046.5: Institutions need privacy, but regulatory black box.

2054.1 --> 2068.9: Yeah, so they're looking at institutional based tokenized workflows, different wallet integration with privacy pools, complian

ce proof SDK, that could be something interesting for begin to think about.

2072.8 --> 2075.8: And then, of course, association set management.

2090.0 --> 2105.6: So I don't think this, you know, this project itself will directly lead to any contribution. This RFP is like expecting, but maybe this project will give some clue.

2106.5 --> 2108.4: Where to start thinking about this.

2109.0 --> 2119.9: But still, like, I mean, as I said, I'm not very sure what they mean by this and especially about the institutional use case of the privacy pools.

2120.8 --> 2127.2: Is that, like, you know, do they mean like it's within the institution or institution as a user of privacy pools?

2128.4 --> 2130.0: That's one question.

2131.2 --> 2145.1: The way I approach that question is they want privacy pools, but with entities with like larger pools of funds to use the technology as opposed to it being individuals using.

2145.6 --> 2150.5: That's kind of how I see the separation between that.

2151.6 --> 2152.7: Size of the fund?

2153.5 --> 2156.9: Well, yeah, just the size of the entity using the protocol.

2156.9 --> 2170.9: I think it might also refer to just some sort of framework that's acceptable to larger institutions like traditional finance institutions.

2170.9 --> 2180.8: So knowing where that associations that comes from is potentially more important than if you're like a D5 protocol kind of thing.

2191.9 --> 2196.7: So if we wanted to target this research at this RFP.

2196.7 --> 2199.2: What would people suggest?

2199.8 --> 2206.8: Would it be mostly around the association set management with the implication that watches are important?

2208.8 --> 2209.8: Do you think?

2219.1 --> 2220.3: Well, yeah, maybe.

2220.7 --> 2221.1: Yeah.

2221.3 --> 2225.5: How institution define clean sets update mechanism and governance.

2225.6 --> 2226.6: Maybe this aspect.

2227.0 --> 2227.4: Yeah.

2227.6 --> 2227.8: Yeah.

2229.2 --> 2233.9: I think that's the most specific focus we could narrow in on.

2236.2 --> 2247.4: Yeah, I mean, this is actually this like how institution define clean sets that's remained ambiguous in the current implementation in my understanding.

2247.8 --> 2261.6: Like, as I say, the Oxbow, I hope I'm pronouncing correctly, like Oxbow has published, you know, working version of the privacy pools, but like, I'm not very sure.

2261.6 --> 2266.9: And I think they didn't disclose like what they're doing.

ing as associations provider.

2267.7 --> 2275.6: So any deposit, it seems like any deposit after waiting for seven days eventually got into the association set.

2275.8 --> 2282.8: And within that seven days, they are doing the like some sort of provenance legitimacy check.

2283.1 --> 2287.6: But again, I don't know what kind of criteria, what metrics they're using.

2287.6 --> 2300.9: And maybe that's not acceptable for institutions like FASP accepting such ambiguous compliance decision.

2301.2 --> 2304.2: Maybe that's why it's not capable.

2304.4 --> 2307.8: I mean, they're saying the institutional use case is not covered yet.

2309.2 --> 2323.6: So a lot of companies like Oxbow or Railgun or any of these kind of privacy pool implementations, what they are doing right now is just buying data from a lot of different blockchain analytics providers.

2323.9 --> 2326.2: And that's kind of how they define their clean sets.

2326.2 --> 2334.6: Whether or not that's acceptable for this particular use case, I would say it probably is.

2334.7 --> 2346.9: But generally their approach is to kind of get as much data from as many sources as possible so that they have kind of the maximum level of visibility into what constitutes a clean association set.

2347.6 --> 2348.8: Okay, I get it.

2351.0 --> 2360.4: Yeah, I knew only one implementation from Oxbow. It's kind of new to me. There are multiple implementations already.

2361.6 --> 2365.7: Okay, thank you for that. That's really important.

2365.7 --> 2386.4: Well, that said, even if they're buying as much data as possible, their criteria for a clean deposit or a dirty deposit is still ambiguous.

2386.9 --> 2391.8: I think the problem still remains, I guess.

2391.8 --> 2395.8: What if...?

2399.2 --> 2413.0: It comes back to the fundamental question, like, you know, what kind of risk assessment or the deposit screening like the other major VASPs like Coinbase are doing.

2413.0 --> 2431.2: I thought they're also using the chain analysis tool, basically buying their result and define by themselves, I mean, define the criteria of the acceptable deposit or acceptable risk by themselves.

2432.2 --> 2434.7: Right, so...

2434.7 --> 2449.1: Yeah, that kind of consensus is totally lacking right now. And basically there's a provider and recipient, right? So provider provides their blacklist or whitelist.

2449.1 --> 2464.9: And maybe they or the depositor make a proof by that information. And after that, they have a recipient to accept the proof, for example, Binance.

2464.9 --> 2482.0: So Binance will do their proof of innocence or anti-money laundering. They need to accept that ZKP, which is proving that it's coming from good addresses or not coming from bad addresses.

2482.0 --> 2499.1: And so how do they make a consensus? So recipient, Binance, provider, somebody, how can they agree on their blacklist or whitelist? This part is totally lacking.

2499.1 --> 2514.6: So that's the reason why none of exchanges are accepting that ZKP right now. So actually, yeah, I'm one of the projects doing this.

2514.6 --> 2532.5: And we are already prepared the mechanism and the proof-of-proving procedure, which people can prove themselves innocent, but still no recipients. Binance is not accepting that.

2532.5 --> 2550.0: Coinbase is not accepting that. It's not only my problem or my project problem. It's all about all projects problem right now.

So who is the first recipient and how to make that?

2550.0 --> 2564.0: And maybe it's one of the biggest problem and obstacle is how to make that consensus. Yeah. So the first step hasn't made yet right now.

2567.8 --> 2569.3: That makes sense so much.

2594.4 --> 2609.2: list, not others. Some might care about what OFSI says in the UK, some might not. So kind of coming to a consensus around what is clean, what is clean enough. I think there's kind of not really this dichotomy.

2609.2 --> 2624.7: Most of the kind of risk scoring mechanisms in our tool and other tools are not black and white. It's all sort of whitelist, blacklist. It's degrees of risk and each company decides their own risk tolerance.

2624.7 --> 2643.9: But part of the problem with a privacy pools based solution is that you are kind of imposing this risk decision on other people. And so right now it's kind of up to Binance or Coinbase to decide, okay, do I trust what Oxbow's association set says?

2643.9 --> 2655.0: Do I trust that anything that's coming from there is okay to accept at my exchange? But then again, if they don't have all of that information, then they might not be able to make that risk determination as well themselves.

2660.0 --> 2683.2: How about incentives? How about the incentives about on the recipient side? Maybe you talk about, yeah, there are different risk tolerance, maybe expectation, but at the same time, most of the recipients here, yeah, some kind of risk taker at the same time they are a business entity.

2684.2 --> 2699.6: How can we make the actual incentives for them to accept this kind of proof of innocence? This is another point of view to tackle.

2699.6 --> 2729.3: One thing that just came to mind when you were saying that is maybe it's the throughput or the frequency of transactions that create the incentive. And I was thinking about when you were speaking before, the problems that you're describing is a centralized provider's perception of a decentralized-ish solution for privacy.

2729.6 --> 2755.2: Which involves CKPs. But if we were only working in the on-chain decentralized world, we focused instead on getting throughput

and having AI agents as the thing of which the association set is either enforced by a strict constraint on an AI agent.

2755.2 --> 2783.4: Or we use the AI as the thing that funnels the transactions with lots of high frequency through these pools, which then adds weight to when centralized providers realize that the volume of transactions on chain is going from 0.5% is going through privacy pools to 5%, to 10%, to 15%.

2783.4 --> 2795.9: And this is scalable and it will scale much more quickly because it's of greater importance to an AI agent to get privacy on chain because their strategies are transparent.

2798.3 --> 2825.2: Yeah, anyway, that's an idea. Another idea is that I've seen an implementation on the AI agent side of things as well, is that you create a privacy pool, which has within it like the X402 payments, which means that you can do private payments within one pool for AI agents, which solves a lot of problems.

2827.3 --> 2839.5: Like the merchant style problems that appear when an agent is operating on your behalf and purchasing a good or service that accepts on-chain finance.

2839.5 --> 2853.3: If the good and service and the agent sit within the same pool and the association set is compliant, then you can get those high frequency transactions within the pool.

2853.3 --> 2859.5: And a transaction within a pool also creates opportunity for efficiency as well, right?

2859.5 --> 2876.8: So, yeah, anyway, some ideas of like re-scoping the go-to-market or privacy pools there to try and get that attention and not relying on this like top-down centralized view of this is what makes a good association set.

2876.8 --> 2893.6: Well, this is what makes a good ZKP, but just like bottom-up, we're all getting this benefit, we're getting this incentive that's happening because of the high frequency and you don't know anything about it because it's within the pool, but you're more than welcome to join.

2894.7 --> 2896.1: Just follow the rules.

2903.5 --> 2905.4: Let me say something.

2905.9 --> 2913.1: So, about the incentive mechanism, I think we can learn from the existing finance, I mean the traditional finance.

2914.3 --> 2926.1: As you know, I'm working in the banking industry, but we offer different interest rates to each customer, so it totally depends on their credibility.

2926.1 --> 2941.0: So, maybe in that case, the customer should be having incentive to maintain their better credit score to get a better interest rate.

2941.0 --> 2967.7: So, but the problem is that that score is totally depends on the each financial institutions, but I think that there's already same issue in the existing finance because especially in Japan, how to calculate credit score is not disclosed publicly.

2967.7 --> 2975.8: So, maybe we need to accept that kind of dilemma here

to calculate trust score.

2979.1 --> 2995.5: Actually, what's interesting is you could build the interest rate based on at the moment of redemption out of the pool, and this is creating an assumption that the pool earns fees because there is some sort of transactions going on inside.

2997.1 --> 3012.2: The interest rate could be informed by how many transaction fees or how good the pool is, in other words.

3012.2 --> 3032.4: And then another thought I had is maybe it is just exchanges as ASP providers, and the exchange ASP is if you want to get privacy before depositing into our centralized exchange go through our particular privacy pool implementation.

3032.9 --> 3036.8: Now, they're not going to go for that, but it would be interesting to see that.

3039.6 --> 3057.6: In that case, the association set provider should be on the association set. I mean, again, I mean, yeah, so this is kind of that almost same as a same idea as the internet. Yeah, the accreditation

3058.4 --> 3059.7: I forgot the name of it.

3061.3 --> 3071.8: Your certification is provided by some certifier and that certifier should have the certification. So, there's always some. Yeah, yeah.

3073.1 --> 3075.2: Exactly, exactly, level of trust.

3082.2 --> 3085.5: The problem. Yeah, as you mentioned that the

3087.0 --> 3097.9: there's always a problem that even if you believed in someone who provided association set is trustful, but on the

3099.1 --> 3110.9: someone you're going to transact with is not believing that person. In that case, even if you are willing to be good behavior,

3110.9 --> 3126.2: you are betrayed by the someone. So, in that case, the someone, I mean the some cryptic, for example, cryptic exchange should be

3126.2 --> 3137.0: should be announcing that the which association sets provider they're going to refer to. Yeah.

3138.9 --> 3139.3: Before

3140.9 --> 3142.8: doing their business, I think.

3143.8 --> 3146.3: Also, they need several

3147.3 --> 3159.1: associates provider, right? Otherwise, no associates provider providing everything so they can know their actual deposit address.

3160.3 --> 3172.4: So, it's hard to make, somehow hard to make sets of associated providers to obfuscate each other.

3174.6 --> 3176.8: Maybe that's another

3179.1 --> 3182.0: problem where we talk about their associate provider.

3182.9 --> 3191.3: Yeah, I was thinking about that around creating, it's almost like a treaty between two privacy pool association sets where

3192.0 --> 3198.4: this treaty outlines what we think is an overlap between our checks and what we think are gaps.

3199.4 --> 3212.4: And then that treaty between privacy pools or association sets can be published. So, it's like very clear when the overlap is

good and when the overlap kicks different deposits out.

3216.0 --> 3225.9: Yeah, the other question is what happens if your funds are deemed clean when you enter the association set, but then obviously things change quite quickly.

3226.2 --> 3233.4: Perhaps your data provider didn't have something labeled as sanctioned then. And then an hour later, it was.

3234.5 --> 3244.9: How do you ensure that, how do you do kind of continuous monitoring to ensure that the association set stays as clean as it was from the outset?

3245.9 --> 3250.6: I actually think proof of honest trajectory is an answer to that question.

3250.9 --> 3258.0: Because you already know that the withdrawal address, because it's coming from the privacy pools withdrawal address.

3258.2 --> 3264.1: So, the separation isn't broken if you issue a proof of trajectory.

3264.1 --> 3275.6: And that means that when you withdraw, the proof of trajectory doesn't, it just almost resets the proof in a way.

3278.8 --> 3287.1: I think the point is, how can we ensure the level of cleanness in the privacy pool? That's the specific association set.

3287.1 --> 3302.4: So, that eventually comes to the governance model, how the association set provider keeps updating the information and how often they actually fetch the information from the provider or something like that.

3302.4 --> 3325.6: And well, if we follow some other model, like establishing, for example, I'm not the professional in this field, but for example in PKI, if I understand correctly, all CA, especially the root CA, they are all subject to the periodic audit by third party.

3325.6 --> 3335.6: And if we really want to ensure the operation of the association set provider, then some sort of audit will be required. That's my understanding.

3338.3 --> 3355.7: Unless everything's calculated in public and cryptographically verifiable, but that's not going to happen in this threat analysis, because some confidential information, optional information, those are all included in the risk scoring.

3356.7 --> 3356.8: Anyway.

3361.7 --> 3379.2: What number do you expect to be kind of enough for the blacklist or whitelist? No, no, a whitelist. Hundred? Thousand? Do you have any actual number?

3381.9 --> 3399.2: If I remember correctly, like for example, the OFAC sanction list got like, maybe there should be someone no better than me, but wasn't it like more than 10,000? So, that's the number of blacklists, right?

3399.2 --> 3417.2: The problem with the labeled public sanctions list is that they don't label every single address that belongs to a particular company that is sanctioned. So, it's not usually really sufficient to just follow that.

3417.2 --> 3441.8: I know there's like free lists out there that just ha

ve the OFAC sanctioned addresses, for example, but for context, like I think they labeled maybe 15 addresses belonging to Garentex, but in our tools, we had millions labeled because they were constantly shifting their wallet structure.

3441.8 --> 3446.8: So, that's another issue that needs to be taken into consideration.

3454.8 --> 3457.6: Are you asking about like a privacy shelling point?

3459.7 --> 3482.3: Basically, I'm asking how many whitelists do we need? Because, for example, there's a thousand blacklists, and I proved that this address is not one of that. Maybe we need to export something like, just a moment.

3485.6 --> 3503.4: Maybe, yeah, we don't have to put our own address out to actual public input to that ZKB circuit. So, maybe it can be private input, right? So, yeah, it must be fine.

3503.4 --> 3527.9: So, maybe if we can prove that, you know, exclusion proof of 10,000 of blacklist addresses, maybe, yeah, it would be fine. And maybe that doesn't matter when it comes to ZKB proving time, like, you know, 10 seconds or one minute.

3529.1 --> 3532.8: So, that's my kind of worry.

3532.8 --> 3554.1: There's two ways to it, though, right? Like, there's the proving that you're not on the list, so the, like, range non-membership proof. But there's also, what I thought you were asking is, like, what is the number that needs to be within the pool of deposits for me to actually get a benefit from joining the pool in the first place?

3554.1 --> 3569.2: And I don't know what that number is. I'm really interested in what that number could be. I think it is actually a ratio more than a specific number. It's like, yeah, open question.

3569.2 --> 3590.2: Yeah, it's just quite depends on how good the tracer is. And in the age of AI, it's getting more and more efficient to trace addresses. So, I think that it's much more than we expect, maybe.

3590.2 --> 3613.5: So, for example, if we have five candidates, we get mixed with five addresses in the same deposit number, deposit amount, maybe five is not enough because each tracers can have many information to tell this is not you or this is you.

3613.5 --> 3626.0: So, maybe, yeah, it's much, much more than we expected, you know, before 2022. And yeah, I don't know, actually.

3626.0 --> 3641.0: So, I think it mostly depends on how unique the user's behavior is in terms of kind of how they deposit into the privacy pool, the amounts that they deposit into the privacy pool.

3643.5 --> 3664.9: Ultimately, the more people who are behaving similarly in the privacy pool, the harder it would be to trace through. But if someone, you know, if everyone else is depositing \$5 and then suddenly someone's depositing a million dollars, then it becomes very obvious which withdrawal from the privacy pool is connected to that.

3664.9 --> 3682.4: And there's a lot of other kind of user behavior mistakes that can allow those linkages to be made. So, you do have to account for kind of user error when you're actually calculating what the realist

ic anonymity set is.

3685.1 --> 3694.7: So, do you think it would be beneficial if there was an agent transformer that you could send your money to then standardize the style of deposit that goes in?

3695.3 --> 3715.1: So, if I had 1 ETH that I wanted to privacy pool and the pool is a 0.1, I would send the 1 ETH to the agent transformer and then that agent would, over time, do 0.1 at a more standard and common pattern.

3718.3 --> 3738.8: But it would depend how many people were using that versus other agents. Because then you would kind of be able to then say, okay, so I know that this person uses this agent to deposit into this privacy pool. So, now I know that anything that matches that agent's pattern on the other side could potentially be one of their withdrawals.

3738.8 --> 3757.1: So, it kind of comes back to the same point of the more people and the more people who are behaving in a standardized way. So, yeah, that does help to an extent. But then you're able to kind of still link the behavior in that they're using that particular agent.

3757.1 --> 3781.3: Which I think, particularly with agentic behavior, those patterns are going to be even more obvious to someone who is trying to trace there. But it's similarly to how, you know, if we're looking at things going into and out of Tornado Cash, for example, like if you see that at a particular gas fee limit is set, you might know what wallet someone is using.

3781.3 --> 3793.4: And that doesn't tell you everything. But if you see that, you know, it's MetaMask on this side and MetaMask on the other side, then that's one more piece of evidence that you have that they could be linked.

3795.1 --> 3813.2: But we can make another question, maybe, because their conclusion can be like open ledger without privacy. No, we cannot use that anymore. No, in this AI era. No, it's possible, right?

3813.2 --> 3837.9: For example, Ethereum-based Bitcoin is an open ledger and Monero, GKS is a private ledger. And maybe public ledger, it should be no longer possible to use when it comes to privacy. It's another possibility.

3838.9 --> 3856.4: No, it just occurred to me because, you know, you guys were talking about AI agents and obfuscation. And yeah, if it doesn't work, people cannot do that, right, because AI is better than people.

3856.4 --> 3868.1: And so if it doesn't work, maybe it's impossible to have privacy on public ledger like that.

3868.1 --> 3885.8: I think you're right to an extent. If it doesn't work, maybe it shouldn't work. And the transparent ledger should be transparent. And we should all realize that that is what a transparent triple entry blockchain does. But I think you need to have the option.

3885.8 --> 3907.3: One slight correction is like Zcash has the transparent and the shielded ledger with that ability to move between, which I like. That's one of the things I like most about Zcash is that you've got that transition in the middle and there's heaps of different things that you

u can do with that.

3908.2 --> 3926.9: But yeah, I kind of like fully agree with your sentiment that we should just realize that transparent blockchains are transparent blockchains and maybe we should not try and retrofit privacy into them and just build private ledgers instead and then actually know which one is good for what.

3926.9 --> 3948.5: And then I think the follow on from that is you then rely on bridges and bridges is this like centralized risk. But if instead you can create like what Zcash has done, this ledger duality with a bit of zero knowledge magic in the middle. Yeah, it's good. It's interesting thought.

3962.6 --> 3977.7: Maybe it's transparent chains, you can only ever get privacy through obfuscation and private chains, you can get privacy through zero knowledge. And that's the education that we should work on.

3980.0 --> 3983.8: Yeah, there's obfuscation as one technique for privacy.

3989.1 --> 3990.9: I'm sorry, go ahead.

3990.9 --> 4016.8: I was just going to say, I think there's still a benefit to privacy, even if it's not perfect privacy. I think trying to achieve this perfectly private ideal, potentially on public blockchains is unrealistic, but some privacy is still better than complete transparency in all situations, depending on why you need the privacy.

4016.8 --> 4041.8: I just meant to say like, yeah, it won't be perfect, of course. Maybe it's not the problem if it's non-perfect or not. It can be possible to kind of meaningless to try obfuscate something on chain because their tracers are getting better.

4042.4 --> 4053.3: They can, yeah, correct many, many information. And, you know, you are talking about their users' mistakes. People can, the tracers can trace everything from there.

4054.3 --> 4070.3: If they are good enough to review anything so that even AI agent obfuscator cannot behave well, not to be traced, it's almost possible to have privacy.

4071.3 --> 4096.6: And only one tracer can destroy the other privacy on chain. It's almost, yeah, you know, impossible. So, yeah, it's just indicating one possibility that we are on the whole wrong track. And, yeah, that's what I want to think about. Not the conclusion, of course, but I want to think about that.

4098.3 --> 4111.2: I also think, though, just sort of playing devil's advocate a little bit, is that in kind of for everyday privacy users, probably no one is going to attempt to trace you through a privacy pool.

4111.2 --> 4127.0: I think there's not that many people who are doing those things and probably they're not going to be focused on kind of smaller sort of everyday use cases generally, I'd say.

4127.0 --> 4149.2: Obviously, that doesn't negate the need for better privacy, but if we're kind of trying to come up with sort of a minimum viable practical implementation that provides sufficient privacy while also preserving compliance, then maybe some tradeoffs are acceptable.

4157.0 --> 4177.2: I feel like it always comes to the question of what kind of privacy and what's the privacy we want in this context? Like, do we need the linkability between the transactions or do we need the, I don't know if it's the appropriate word, but the complete privacy, even including the option identities?

4177.2 --> 4191.8: I mean, well, I think someone brought up the point like how much extend or how much privacy we want, but that's tolerated. I mean, that's acceptable.

4191.8 --> 4203.0: But I mean, before going to that point, like maybe some sort of differentiation of the terminology will be maybe required. That's what I felt.

4204.4 --> 4223.4: And, I mean, in my understanding, the mixing services like torrenting the cache, privacy pool, that's completely focusing on like, I mean, as I say, cutting the linkability between the transactions, right?

4223.4 --> 4244.4: And, I mean, what I wanted to say is, and well, so I think at least privacy pool has succeeded in defining a different perspective of, oh, I'm sorry, a different level of the privacy.

4244.4 --> 4266.7: Like, for example, torrenting the cache, in torrenting the cache, or even in zcache, like the whole pool, in my understanding, and the whole pool, the whole transaction space is going to be anonymity set, but by creating association set, we risk, we, you know, sacrifice some sort of privacy by narrowing down the association set.

4266.7 --> 4279.7: Still, it's going to, how to say, guarantee some sort of compliance, supposed to be guaranteeing the privacy, I'm sorry, the compliance level within the pool.

4279.7 --> 4302.1: And so that's where the trade off is, and I don't really think this direction is wrong, or this is not the right track, but still, well, as this RFP clarifies, like there are so many questions, I mean, so there are so many like missing parts and questions to solve.

4304.2 --> 4306.1: But yeah, that's my random thought.

4308.4 --> 4309.7: Not even a question.

4328.4 --> 4329.4: Okay.

4329.9 --> 4334.0: That feels like it's met a natural end.

4335.3 --> 4337.2: Unless someone wants to

4343.6 --> 4344.9: Okay, cool.

4346.4 --> 4352.4: So in terms of the this session, we really mostly wanted to focus on privacy pools.

4353.1 --> 4356.7: There is also like it's titled accountable wallet.

4357.6 --> 4360.6: And there were some other, wait, let me click on it.

4365.7 --> 4370.3: I'm just going to go through the agenda to make sure we didn't miss anything that we said we would cover.

4371.7 --> 4390.6: We wanted to define accountable wallet. I think we, we, we didn't go there because we're focusing on the privacy pools, but there was another point on verification data records, which is similar to what you were saying, Keegan, around V, C, A, or V.

4393.4 --> 4406.9: Another thing that I have been working on recently, w

high I can share with the group is recovery through proof of understanding of keys.

4410.4 --> 4412.9: So how do I describe this?

4412.9 --> 4430.6: So when using AI and LLMs, you can have them compress like large documents into specific sentences, which are always going to be different.

4431.2 --> 4451.9: However, mirror and the word proverb comes to mind here. And why I put that in the agenda today is this new technique and it uses that I want to share with the group that I, it uses the Zcash ledger duality that I was mentioning.

4451.9 --> 4476.3: So what you do is someone reads your work. They put your work into an LLM within that signal. There's a prompt instruction, which has the LLM create a proverb, which is a summary in two sentences of the signal, which is the whole document.

4476.3 --> 4490.1: And for that to then be a proof of understanding that someone sends to you on the shielded ledger side of Zcash in an encrypted memo attached to a transaction.

4491.0 --> 4501.2: And when I receive that shielded transaction, it's a signal that someone has understood my work and would like to provide a proof of understanding.

4501.2 --> 4519.8: So what I do is then make sure that that compression of the document matches, work out which document or which signal they're matching. And I can do that very easily because I would just match the proverbs semantically.

4519.8 --> 4537.6: You can get an LLM to do that to work out which full reconstruction they're talking about. And then you publish the proverb in a hex in the same way that ordinals happen on Bitcoin.

4537.6 --> 4550.7: But what you're doing is you're actually deshielding the shielded memo that someone sent to you as a discovery mechanism that they can then selectively disclose that they were the one to send you the shield memo.

4550.7 --> 4557.6: So you're actually separating a bilateral proof of understanding using Zcash as your knowledge layer in the middle.

4558.6 --> 4576.4: And I think that that is quite novel and fun because it uses human readable language as the proof while it's less secure than using the sort of map.

4577.6 --> 4599.4: It gets on Zuko's triangle, it gets closer to the top of the triangle, like the human meaningful side rather than the decentralized and secure side, which could be a good alternative for proving that you understood something about someone and then that person received that and then publish that so you can become discoverable but unlinkable.

4600.4 --> 4616.4: So that's something that I have built. It's a combination of two things. One is like an oracle that interacts with Zcash and the other is the compression system and the relationship proverb protocol.

4617.1 --> 4624.6: So happy to share more on that. The article that I wrote is called Understanding as Key.

4626.3 --> 4647.9: And the exploration that I went down with Understanding

ng as Key is because it's words that are human meaningful, which are the thing that's getting put within a hash or within a shard that you put on chain as an inscription or ordinal.

4648.9 --> 4656.1: You can actually play with the visibility ratio, which then becomes a representation of the relationship between two people.

4656.1 --> 4668.5: So say, for example, I want Leona in my relationship. He sends me a proof of understanding and we are 50-50. We're like a fair relationship.

4669.1 --> 4678.6: I could get the first half of his proverb and the second half of my proverb and make that the visible thing, words in the ordinal.

4678.6 --> 4692.7: But we both actually are what formed that key as an inscription on chain. So we can both recover the whole thing.

4694.5 --> 4705.3: Leona could recover my half and I could recover his half for him in terms of just selectively disclosing that we did this thing and we wanted to provide a proof on a blockchain about it.

4706.3 --> 4711.9: And that ratio actually is really interesting because it represents a relationship.

4712.5 --> 4729.7: Because if I was, say, a public organization and happy for my proverb to be fully visible, then I accept inputs of people who agree with my manifesto, with my document.

4730.3 --> 4738.6: They send the shielded memos. A whole bunch of people are agreeing and we just keep publishing our side of the proverb.

4738.9 --> 4752.9: But the hash, the shah that is attached to it has all of these people who agree with that manifesto as a discoverability mechanism and a recovery mechanism or selective disclosure mechanism for them to be like,

4752.9 --> 4760.4: yes, I signed the manifesto on this date and I can prove that because I have the shielded memo if I ever want to give someone a viewing key.

4762.5 --> 4771.5: So I thought that was quite cool and fun and novel. So I'd like to propose it as a discussion point for the group today in our last half an hour.

4774.1 --> 4784.5: It's similar to privacy pools, but we're talking about human meaningful side rather than money in terms of value.

4796.4 --> 4812.3: So cybertech is on chain and we can submit a key on chain so that people can see this is a decrypted techist.

4819.1 --> 4823.5: Is there any similar case before?

4824.5 --> 4827.0: I don't know. This is why I'm asking people.

4828.7 --> 4833.2: I've never heard of that. It's quite new.

4834.2 --> 4841.2: Only one similar case I remember was like re-cyphering.

4843.6 --> 4867.1: It was like cybertech is on chain and kind of re-cypher, meaning that converting that to another cybertechist, to another person, based on their authorization or key condition.

4868.8 --> 4879.6: It doesn't represent any share. So yeah, in that sense, it's nothing similar, but kind of on chain encryption governance.

4880.6 --> 4893.5: Maybe to implement that, maybe most easiest and interesting way is using VET key, VET key of ICP.

4894.6 --> 4905.8: I think it's really interesting because basically VET key is like a decentralized key management.

4905.8 --> 4918.8: So maybe it's more like leaked protocol. Leaked protocol is using TE and the VET key is based on their ID base encryption.

4921.5 --> 4931.4: Actually for me, it's only one case ID base signature is being used for the real world purposes.

4932.7 --> 4935.9: That's the reason why I got quite excited.

4936.8 --> 4944.2: But yeah, I think that to make that kind of encryption management, I think that the VET key is the best idea.

4944.2 --> 4949.2: It's also on chain and also, yeah, it's flexible.

4950.9 --> 4970.0: So maybe if we use that, if we implement your idea with kind of on chain key submission, maybe it's a bit difficult to make a new key maybe.

4970.4 --> 4972.9: That's the only difference we need to care.

4976.5 --> 4984.8: Where I was thinking about that could expand is that you would need more than one instance.

4986.0 --> 4997.5: So if you and I did 10 of these proof of understanding, like bilateral ceremonies on chain, then what if we got one word from all of them?

4997.5 --> 5003.5: And then, oh, let's say we did 12 together and we got one word from all of them.

5003.6 --> 5014.5: That is actually a, it's a bilateral key ceremony in a way that is on chain recoverable.

5015.8 --> 5021.8: And then based on all of the ratios of all of the different ones that we did.

5021.8 --> 5027.8: So maybe for half of them, you revealed 50% of yours and half of them, I revealed 50% of mine.

5028.0 --> 5031.5: Or maybe there's a couple of them in there that we wanted full responsibility.

5031.9 --> 5035.0: So we did a 90% or an 80% ratio.

5036.7 --> 5039.2: It's really hard to work out what the ratio is, right?

5039.2 --> 5046.8: Because they're words, they're matching proverbs in a way, like they're matching meaning.

5046.8 --> 5051.8: But it's the, like your context is what creates the meaning.

5051.8 --> 5056.8: And my context is what creates the meaning with the like base information.

5061.0 --> 5074.0: So, yeah, I think it's potentially gets us to this like social key recovery, like where you can almost create.

5074.0 --> 5090.6: And I remember like Vitalik was talking about this a little bit, where you create these like social key recovery pools of like everyone in your Harvard class can help you recover a key because they all have the same proof.

5092.0 --> 5098.4: Only it's organic and bottom up and it's not top down

credential to create the pool.

5098.5 --> 5103.0: It's anyone who understands the same thing and is able to compress that meaning.

5114.9 --> 5132.3: So maybe the use case, no implementation of that idea, maybe it's urgent wallet similar to that idea, like no guardian recovery.

5133.3 --> 5154.6: What do you think about kind of recovering the actual key cryptographic way on-chain or off-chain or just to kind of multi-sig or kind of account abstraction without the off-chain cryptography?

5155.1 --> 5156.9: Which is the best way do you think?

5156.9 --> 5163.3: I think I was thinking account abstraction originally, which is kind of the same as the multi-sig.

5166.1 --> 5172.9: But, yeah, I hadn't thought about doing the cryptography on-chain before today.

5173.3 --> 5175.4: So thanks for letting me know about that.

5176.1 --> 5176.7: That's interesting.

5184.7 --> 5193.1: Yeah, you can create like guardian pools as being the source of the signal truth.

5194.4 --> 5202.8: And perhaps actually this leads to the watches in a way and the association set providers.

5203.1 --> 5217.1: If the association set provider publicly discloses a very clear framework of this is what our association set does and in order to be a part of this pool you need to go through this key ceremony.

5217.4 --> 5228.6: Then that adds another layer of trust to the association set.

5229.2 --> 5243.0: And also to the person who is submitting the deposit to the association set, like it's a bilateral proof of understanding that this is what I'm doing and this is the association set provider that I'm working with.

5243.0 --> 5249.6: It's not I'm an association set provider and these are my terms and I'm going to impose them on you.

5249.6 --> 5251.8: It kind of flips that a little bit.

5254.5 --> 5262.8: Yeah, but these are all expansions on the idea that we need to make simple and succinct.

5265.4 --> 5284.2: However, I think about the biometric key generation like, you know, recently Hitachi is doing that and somehow a deterministic way of the key generation from biometrics.

5284.2 --> 5292.5: Maybe most of the cases it's recovering some key which is lost by some mistake, isn't it?

5295.8 --> 5298.7: Yeah, but I shut up when I think about biometrics on chain.

5299.9 --> 5303.8: Yeah, you know, it doesn't have to be on chain.

5305.9 --> 5314.6: Sorry, I just arrived. Is it still about the accountable wallet topic with biometric one or it's you?

5315.8 --> 5319.1: So we spoke mostly about privacy pools in this conversation.

5319.1 --> 5330.9: But I just shared the understanding as key research that I've done recently where you use these like proverb compressions as t

he like key ceremony.

5331.4 --> 5337.2: And the way that you do that is you share the proverb through an encrypted channel on chain.

5337.2 --> 5349.2: So through a shielded memo and then one of the people who the person who receives that can selectively disclose that to the transparent chain through like an ordinal style inscription.

5350.1 --> 5356.7: So what was your question about biometric based wallet related to this topic or another one? I was not sure.

5357.6 --> 5371.7: So basically the key recovery, the pool for the key recovery, if we can eliminate that for by some reason, yeah, it'd be kind of helpful, I thought.

5372.5 --> 5379.8: So we published a document some time ago about the different kind of wallets.

5379.9 --> 5392.0: So there is a discussion around the use of biometrics for either generating or binding the wallet, generating a key or binding the wallet to a person.

5392.7 --> 5398.3: There was a presentation made also at the last block, 13, I guess.

5398.3 --> 5407.8: So there are some options to do it, but there are also some cons when doing this.

5410.0 --> 5420.2: And I should mention also that when you start discussing with the Web3 and crypto community about the use of biometrics for wallets,

5420.2 --> 5429.8: there are kind of a strong pushback because of course this is not seen as something which is really in the spirit of the crypto developers.

5430.7 --> 5437.1: But definitely there are things to be investigated and yeah, Hitachi is one of the players on that side.

5437.6 --> 5442.2: NEC as well, yeah, and I think this was NEC team who made the presentation in block 13.

5442.2 --> 5442.9: Yeah.

5449.2 --> 5463.0: When it comes to biometrics key, in my opinion, the backup is 100% required because like you said, the biometrics could be not, yeah.

5463.0 --> 5473.1: So that's why biometrics is good, but at the same time, my concern is about like needing the backup.

5485.8 --> 5491.2: So from my shallow understanding from the biometrics wallet paper, right.

5491.2 --> 5506.4: And my understanding is that you actually don't have to back up the biometric or the secure sketch or fuzzy extractor or something like that because you can always regenerate from.

5507.4 --> 5510.9: And you only need to store the help of it or something like that, right.

5511.2 --> 5512.3: You are the experts.

5512.8 --> 5513.3: Yeah.

5514.0 --> 5518.8: Well, I've been working on this topic for quite some time.

5519.9 --> 5520.8: Yeah, indeed.

5520.9 --> 5530.2: You have ways to basically correct the errors, what you can consider as errors from one capture to another one.

5531.2 --> 5550.9: So by using some cryptographic techniques, error correcting techniques, you are able to reconcile basically the differences between two captures of the same person, of the same biometric source and to generate the same key.

5552.5 --> 5564.7: Well, with one exception, if you use one finger and your finger is cut by accident, hopefully by accident, only by accident, then you cannot do much.

5565.1 --> 5567.5: So in that case, you need the kind of backup, of course.

5574.9 --> 5582.7: I guess a lot of pushback towards that kind of technology is because of perception, not a real thing, right.

5589.8 --> 5603.1: I think part of the pushback is also the organizations that use it as first cause or like first use case.

5603.1 --> 5615.4: And especially something I've been observing in the UK around that debate at the moment is the pushback comes from, do we have to start with this?

5617.1 --> 5617.9: Yeah.

5623.2 --> 5636.3: In general, when we mentioned biometrics for any kind of application, first, one of the first requirements from regulation is you have to make it optional.

5637.8 --> 5652.4: So even if we talk about an application that is not meant necessarily to be in a specific jurisdiction, decentralized applications, let's say, well, in principle, it should be the same.

5652.9 --> 5663.4: Making it optional, make it kind of easier in terms of perception for the end user to accept it because then if you at least you give them the choice.

5667.4 --> 5672.8: So how fuzzy is that extractor? Actually, I don't know.

5674.6 --> 5684.6: It's just awkward to me as a question because, for example, the entropy is not defined when it comes to biometrics.

5684.6 --> 5705.0: So, for example, if I drop one blood and some CIA, Mossad, I don't know, they analyze everything and they reconstruct my finger and make a fingerprint.

5705.4 --> 5709.7: So what's the actual entropy we can have with biometrics?

5709.7 --> 5716.2: Are we hijacking the topic while discussing biometrics or is it okay to continue?

5716.6 --> 5718.6: I don't know whether it was the agenda.

5718.8 --> 5718.9: Okay.

5720.3 --> 5735.9: So there are different ways to measure the entropy and we should not conflate noise with entropy.

5736.7 --> 5737.4: What I mean?

5738.9 --> 5765.6: For a given modality, let's say face, for instance, versus finger, versus irises, when you capture same biometric source, let'

s say your finger, between different periods, under different environment, different pressure on the sensor and so on, and lights can also have an impact.

5765.6 --> 5768.3: You have noise from one capture to another one.

5771.5 --> 5777.7: For, let's say, irises, for instance, you have different ways to measure the noise.

5778.1 --> 5787.1: In general, you measure the noise after extracting some meaningful information that you are going to leverage to be able to make a comparison.

5787.1 --> 5810.8: To give you an example, for irises, you generally represent it as a binary code, which is called an iris code, and you may have to render something like 40% of errors between one capture to another one.

5810.8 --> 5835.2: The thing is that, in fact, you represent an iris as an iris code with something like 2,000 to up to 10,000 bits in order to be able to get rid of the noise in a way, but at the end, what you want to do is to be able to determine whether this is the same person or not.

5835.2 --> 5841.1: To determine whether this is the same person or not, you don't need that many bits at the end.

5841.2 --> 5845.4: We are only a bit more than 7 billion on Earth.

5846.1 --> 5853.7: You don't need many bits to be able to provide a kind of uniqueness for everyone.

5853.7 --> 5883.6: At the end, the entropy of the biometric itself is related to the number of persons that you want to identify in a way, to be able to uniquely identify, which means that the way to see the entropy for biometrics is more on the ratio of errors that you will make while trying to distinguish two persons.

5884.8 --> 5902.6: If you have a kind of bad biometric modality that is not really unique, like the shape of your hand, basically the entropy, let's say maybe the error rates would be 5% to 10%.

5902.6 --> 5905.1: Let's say the entropy is very low.

5912.6 --> 5915.0: It's 3 bits, basically.

5915.6 --> 5924.4: While for iris, if you take one iris or you take two irises, then you have something which starts to be close to 100 bits.

5926.8 --> 5944.4: That's where it's important to distinguish the number of errors that you have, the noise that you have to handle, with respect to the entropy for a specific objective of why you are using a biometric comparison.

5949.4 --> 5951.9: Yeah, that was really good. Thank you.

5952.9 --> 5964.3: One thing that I wrote down was living biology is a non-fixed entropy equation, whereas mathematics could be a fixed entropy equation.

5964.9 --> 5966.8: That's a simple way to put it.

5968.0 --> 5983.1: If you take DNA, for instance, which is kind of biometrics, I'm not saying that we should use DNA for biometrics, but if you look at DNA, there is a lot of noise in DNA.

5983.1 --> 5989.1: But the entropy is very high, because you have tons of

f information.

5989.9 --> 6011.5: But at the end, when you use DNA in a crime investigation or in court, you only use something like 20 to 40 indicators to determine whether this is the same person or not.

6011.5 --> 6022.2: Because the way you measure, the way you extract the information from the DNA is so precise that you are almost sure at some point that this is the same person.

6023.4 --> 6026.6: Or at least within the same family group, in a way.

6027.8 --> 6035.4: How about using your pet's DNA, for example, cats? No body investigates that, right?

6035.4 --> 6037.1: Oh my gosh, did you just say that?

6037.8 --> 6052.6: It makes me think about the authentication means where you are using your own object at home in order to do the recovery process.

6053.0 --> 6060.3: You have specific postcards, you have specific souvenirs that you have at home.

6060.4 --> 6066.0: You are the only one to know them, and you use them in order to combine them to produce a kind of password.

6066.0 --> 6069.6: And this is the same principle as for biometrics.

6069.8 --> 6077.6: At the end, you use techniques like fuzzy extractor, secure sketch, in order to be able to correct some errors.

6077.6 --> 6083.8: Because sometimes you have one object missing, you still want to be able to reproduce a password.

6084.8 --> 6088.3: And with many objects, you may reproduce a key.

6089.1 --> 6093.2: That's a whole new meaning to lifecycle management of keys.

6102.8 --> 6104.9: I think there's a startup there.

6105.9 --> 6114.1: I remember at the very beginning of the first iPhones with the fingerprint sensor on it.

6114.3 --> 6118.6: I think it was the 4S, or maybe the 5S, I don't remember.

6119.4 --> 6124.4: It was quite easy to spoof the sensor.

6125.4 --> 6138.1: So there were some researchers trying to produce fake fingers, for instance, just to be able to use it without registering with their own finger.

6138.2 --> 6139.7: That was the same principle.

6139.7 --> 6149.4: You can generate as many fake fingers as you like, and you can decide which one you want to use in order to authenticate on your iPhone.

6149.9 --> 6151.4: It's kind of the same.

6157.7 --> 6162.0: We have 10 minutes left for this session.

6162.6 --> 6165.0: I think this has been really awesome.

6165.3 --> 6166.9: We've covered some great bases.

6166.9 --> 6176.9: We started with privacy pools, then moved on to understanding as key, and then biometrics for key management as the recovery mechanism, too.

6178.9 --> 6182.9: Are there any other key topics that...

6187.1 --> 6190.2: We have a session on offline key management tomorrow.
6193.8 --> 6195.9: Did you just say that's a keeper?
6202.8 --> 6214.2: Opening it up to the floor, Keegan, did you have anything else that came to mind based on the feedback on your presentation that you wanted to summarize for the group before we close out?
6216.3 --> 6220.1: That was a really insightful discussion, actually.
6220.6 --> 6235.6: I appreciate all the feedbacks I got today, especially the discussion about the RFP from the Ethereum.
6236.6 --> 6240.3: Diving into what the institution will mean.
6242.2 --> 6259.4: The one feedback I really found interesting is how we can incentivize those institutions to accept zero-knowledge proof, like some third-party-generated compliance criteria or something like that.
6260.4 --> 6267.0: Also, referring the TradFi custom is also important, I guess.
6267.6 --> 6276.3: That's the potential direction that we can proceed this project after further clarification of the current stuff.
6276.9 --> 6280.1: I really appreciate it, and thank you so much.
6280.3 --> 6286.9: That's my comment, and I think I basically covered everything I want to talk about today.
6289.4 --> 6290.5: Brilliant, thank you.
6295.4 --> 6309.9: So the next topic for the Block 14 is going to be in 10 minutes, and it will be the whole group in this one room.
6310.3 --> 6315.0: And the topic is going to be forensic analysis towards a common lexicon of harmful on-chain activities.
6315.0 --> 6317.1: See you then.
6373.6 --> 6375.0: Thank you.
6403.6 --> 6405.0: Thank you.
6433.5 --> 6434.9: Thank you.